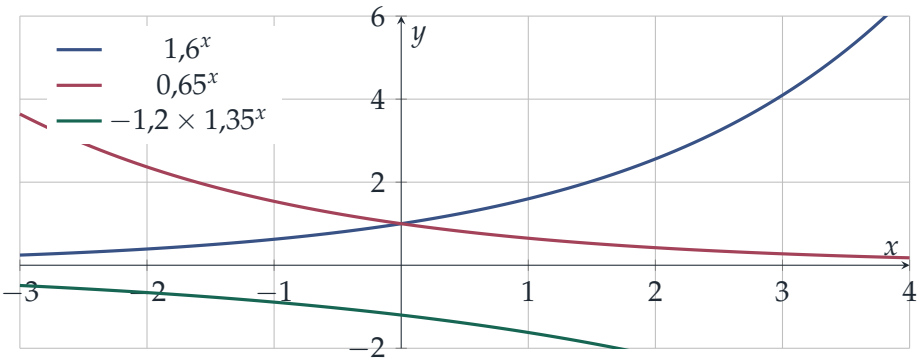
Chapter 2

Exponential functions

Skills: model a continuous change, switch representation, calculate, interpret

2.1 Reference notes

For $a > 0$, the function $x \mapsto a^x$ extends to all real numbers the integer powers of a. It satisfies $a^0 = 1$ and $a^{-x} = \dfrac{1}{a^x}$.



Key ideas

If $a > 1$, $x \mapsto a^x$ is strictly increasing. If $0 < a < 1$, it is strictly decreasing. For $f(x) = ka^x$, multiplying by $k < 0$ reverses the direction of variation. In all cases, $a^x > 0$.

For $a > 0$ and all real numbers x, y :

$$a^{x+y} = a^x a^y, \qquad a^{x-y} = \frac{a^x}{a^y}, \qquad (a^x)^y = a^{xy}.$$

Common mistake

In general, $a^x + a^y \neq a^{x+y}$. The properties concern products and quotients, not sums.

2.1.1 Changes and average rate

A quantity with initial value V_0 subject to a constant relative change with multiplier q is modelled by $V(t) = V_0 q^t$. If an overall multiplier Q results from n identical changes, the average multiplier is $q = Q^{\frac{1}{n}}$ and the average rate is $t = q - 1$.

Worked example

Turnover rises from 80,000 to 95,000 euros in four years. The overall multiplier is $Q = \dfrac{95\,000}{80\,000} = 1{,}1875$. The average annual multiplier is $q = 1{,}1875^{\frac{1}{4}} \approx 1{,}0440$: the mean annual rate

is therefore approximately 4.40 %.

2.1.2 Solve and interpret

When both sides can be written with the same base, we use injectivity: for $a > 0$ and $a \neq 1$, $a^u = a^v$ is equivalent to $u = v$. Otherwise, a graphical or numerical method gives an approximate value; the common logarithm, studied later, will give a formula.

Session 7 — Definition and direction of variation of a^x

Exponential functions

06/10/2026 • B : 2 h – C : 1 h

Objectives of the session

· Give meaning to a^x.

· Link base and variation.

· Read curves.

Fluency · 6 to 8 minutes, without a calculator

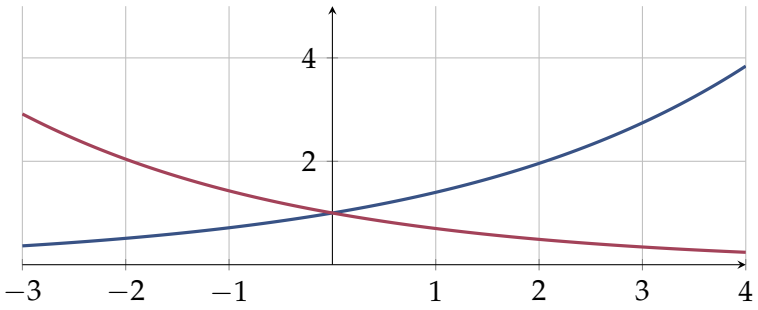
1. Write the multiplier for a 10% increase
2. Compare 1.04^3 and 1.04^7. .....
3. Calculate $2^4 \times 2^3$. .....
4. Write 5^{-2} in fraction form .....
5. Is the function $x \mapsto 0{,}7^x$ increasing? .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: give meaning to a^x and link the base and the variation.

Investigation

Using a calculator, complete a table of values of $1{,}5^x$ and $0{,}7^x$ for $x = -2, -1, 0, 1, 2, 3$.
Plot the points, conjecture the variations and identify the point common to both curves.



Written summary

For $a > 0$, a^x is defined for every real number x. The function is increasing if $a > 1$ and decreasing if $0 < a < 1$.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Simplify $3^{x+4} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 4 \times 1{,}04^x$ and $g(x) = -4 \times 0{,}8^x$.
3. An initial value of 1350 increases by 4% per year. Give the model and calculate the value after 6 years.

References in the textbook

Extensions: Calao, lesson pp. 18-21; exercises p. 24, nos. 1-3. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 48, Exercise 2

On 1st January 2019, France has 66,9 million inhabitants. Its annual growth rate is estimated at 0,4%. For every natural number n, let u_n be the population of France in millions on 1st January of the year $2019 + n$ and we assume that the sequence (u_n) is a geometric sequence.

1. Give the common ratio of the sequence (u_n). Express u_n in terms of n.
2. Suggest an expression for a function f for estimating the number of inhabitants in France x years after 1st January 2019, x being a real number. Determine, according to this model, the number of inhabitants on 1st January 2022, on 1st July 2023 and on 1st October 2025.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 155 and the rate or common ratio with 3. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 8 — Powers and algebraic properties

Exponential functions

08/10/2026 • B : 1 h – C : 2 h

Objectives of the session

· Transform powers.

· Use negative exponents.

· Identify an inapplicable rule.

Fluency · 6 to 8 minutes, without a calculator

1. Simplify $x + 4 - x + 2$. .....
2. Calculate $2^{1} \times 2^{3}$. .....
3. Write 5^{-2} in fraction form .....
4. Is the function $x \mapsto 0{,}7^{x}$ increasing? .....
5. Compare 1.04^{3} and 1.04^{7}. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: transform powers and use negative exponents.

Investigation

Simplify $2^{3} \times 2^{-5}$, $5^{x+2} \div 5^{x}$ and $(3^{x})^{2}$. Then invent a numerical counterexample to $a^{x} + a^{y} = a^{x+y}$.

Written summary

$a^{x+y} = a^{x}a^{y}$, $a^{x-y} = \dfrac{a^{x}}{a^{y}}$ and $(a^{x})^{y} = a^{xy}$. These rules require the same positive base.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Simplify $3^{x+5} \div 3^{x}$ and $(2^{x})^{3}$.
2. Study the direction of variation of $f(x) = 5 \times 1{,}05^{x}$ and $g(x) = -5 \times 0{,}8^{x}$.
3. An initial value of 1400 increases by 5% per year. Give the model and calculate the value after 6 years.

References in the textbook

Extensions: Calao, lesson pp. 18-21; exercises p. 24, nos. 1-3. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 49, Exercise 10

Write as a single power of 2:

$$a = 2^5 \times 2^3; \qquad b = \left(2^{4,5}\right)^3; \qquad c = 2^3 \times 4^2; \qquad d = \frac{2^5}{4}.$$

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 160 and the rate or common ratio with 4. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 9 — Functions $x \mapsto ka^x$: variation and graph

Exponential functions

13/10/2026 • B : 2 h – C : 1 h

Objectives of the session

· Study ka^x.

· Combine sign and variation.

· Link formula and graph.

Fluency · 6 to 8 minutes, without a calculator

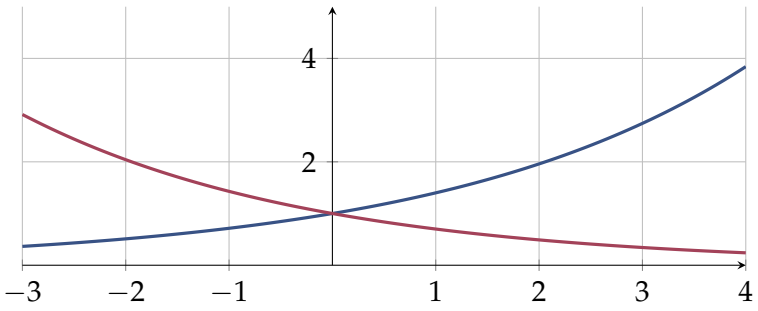
1. Calculate 20 per cent of 100 . . . . . .
2. Write 5^{-2} in fraction form . . . . . .
3. Is the function $x \mapsto 0{,}7^x$ increasing? . . . . . .
4. Compare 1.04^3 and 1.04^7. . . . . .
5. Calculate $2^2 \times 2^3$. . . . . .

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: study ka^x and combine sign and variation.

Investigation

Match the four functions 2^x, $0{,}6^x$, -2^x and $-0{,}6^x$ to four curves. Justify each choice by the sign, the y-intercept and the direction of variation.



Written summary

For $f(x) = ka^x$, the sign is that of k. If $k < 0$, multiplication by k reverses the direction of variation of a^x.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Simplify $3^{x+6} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 6 \times 1{,}06^x$ and $g(x) = -6 \times 0{,}8^x$.
3. An initial value of 1450 increases by 6% per year. Give the model and calculate the value after 6 years.

References in the textbook

Extensions: Calao, lesson pp. 20-21; exercises p. 24, nos. 11-13, then p. 28, nos. 43-44. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 48, Exercise 6

Determine the direction of variation of the functions defined on $\mathbb{R}$ by:

$$a : t \mapsto -2 \times 1{,}4^t ; \qquad b : t \mapsto 9{,}85 \times 0{,}95^t ; \qquad c : t \mapsto 0{,}8 \times 2{,}25^t .$$

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 165 and the rate or common ratio with 5. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 10 — Average rate of change

Exponential functions

15/10/2026 • B : 1 h – C : 2 h

Objectives of the session

· Calculate an average rate.

· Distinguish overall and average rate.

· Interpret an exponent $\frac{1}{n}$.

Fluency · 6 to 8 minutes, without a calculator

1. Write the multiplier for a 25% increase
2. Is the function $x \mapsto 0{,}7^x$ increasing? . . . . . .
3. Compare 1.04^3 and 1.04^7. . . . . .
4. Calculate $2^3 \times 2^3$. . . . . .
5. Write 5^{-2} in fraction form . . . . . .

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: calculate an average rate and distinguish overall and average rates.

Investigation

The price of a service increases from 18 to 23,50 euros in five years. Calculate the overall multiplier, then the average annual multiplier and the average annual rate. Check by compounding the change five times.

Written summary

If Q is the overall multiplier of n identical changes, the average multiplier is $q = Q^{\frac{1}{n}}$ and the average rate is $q - 1$.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Simplify $3^{x+2} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 2 \times 1{,}02^x$ and $g(x) = -2 \times 0{,}8^x$.
3. An initial value of 1500 increases by 2% per year. Give the model and calculate the value after 6 years.

References in the textbook

Extensions: Calao, lesson pp. 20-21; exercises p. 24, nos. 11-13, then p. 28, nos. 43-44. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 49, Exercise 18

A company's turnover increased by 38% over three years. Calculate the average annual increase to the nearest 0.1%.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 170 and the rate or common ratio with 6. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 11 — Reading graphical representations

Exponential functions

03/11/2026 • B : 2 h – C : 1 h

Objectives of the session

· Read images and preimages.

· Compare two models.

· Estimate graphically.

Fluency · 6 to 8 minutes, without a calculator

1. Simplify $3\ x + 7 - x + 2$. .....

2. Compare 1.04^{3} and 1.04^{7}. .....

3. Calculate $2^{4} \times 2^{3}$. .....

4. Write 5^{-2} in fraction form .....

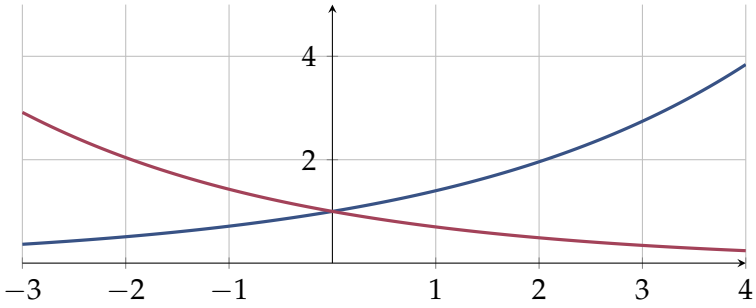
5. Is the function $x \mapsto 0{,}7^{x}$ increasing? .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: read images and preimages and compare two models.

Investigation

Two commercial offers are modelled by $f(x) = 120 \times 1{,}08^{x}$ and $g(x) = 180 \times 0{,}96^{x}$. Draw the curves on $[0, 12]$, read their intersection point and interpret the regions where each value is the larger.



Written summary

Reading $f(x)$ means starting from x to the curve then to the y-axis. Finding a preimage of y means starting from y to the curve then to the x-axis.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Simplify $3^{x+3} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 3 \times 1{,}03^x$ and $g(x) = -3 \times 0{,}8^x$.
3. An initial value of 1550 increases by 3% per year. Give the model and calculate the value after 6 years.

References in the textbook

Extensions: Calao, p. 52, nos. 31 to 36. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 52, Exercise 38

Below are the graphs of the functions f and g defined by: $f(x) = 2{,}9^x$ and $g(x) = 0{,}35^x$.

1. What is the graph of f? Justify this choice.
2. Solve the equation graphically $f(x) = 4$. A graphing calculator can give an approximate solution to within 0.01.
3. Determine graphically the set of real numbers x such that: $f(x) \leqslant 4$ and $g(x) \leqslant 4$.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 175 and the rate or common ratio with 7. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 12 — Simple exponential equations

Exponential functions

05/11/2026 • B : 1 h – C : 2 h

Objectives of the session

· Solve with the same base.

· Approximate a solution.

· Check plausibility.

Fluency · 6 to 8 minutes, without a calculator

1. Calculate 5 per cent of 160 .....
2. Calculate $2^1 \times 2^3$. .....
3. Write 5^{-2} in fraction form .....
4. Is the function $x \mapsto 0{,}7^x$ increasing? .....
5. Compare 1.04^3 and 1.04^7. .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: solve with the same base and approximate a solution.

Investigation

Solve $4^x = 64$, $9^x = \frac{1}{3}$ and $1{,}12^x = 2$: first by using the same base where possible, then numerically for the last one. Bound the last solution between two integers.

Written summary

To solve $a^x = b$: use the same base when possible; otherwise bound or use a digital tool. Check that $b > 0$.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Simplify $3^{x+4} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 4 \times 1{,}04^x$ and $g(x) = -4 \times 0{,}8^x$.
3. An initial value of 1600 increases by 4% per year. Give the model and calculate the value after 6 years.

References in the textbook

Extensions: Calao, p. 52, nos. 37-38, then p. 54, nos. 45-46. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the textbook's digital resources.

Calao textbook · p. 52, Exercise 33

Solve on $\mathbb{R}$ the inequality: $0.8^{x} \geqslant 0{,}8^{3}$.

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 180 and the rate or common ratio with 2. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 13 — Modelling with an exponential function

Exponential functions

10/11/2026 • B : 2 h – C : 1 h

Objectives of the session

· Build an exponential model.

· Identify the parameters.

· Predict and interpret.

Fluency · 6 to 8 minutes, without a calculator

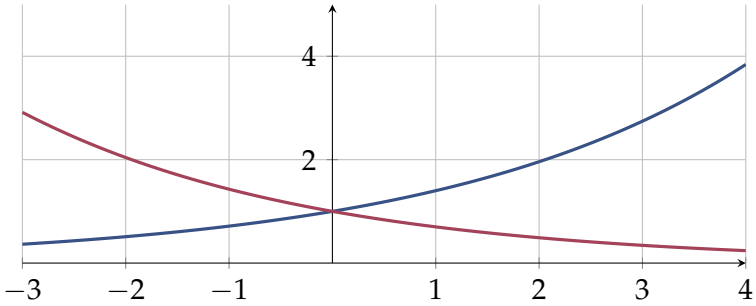
1. Write the multiplier for a 10% increase
2. Write 5^{-2} in fraction form . . . . . .
3. Is the function $x \mapsto 0{,}7^x$ increasing? . . . . . .
4. Compare 1.04^3 and 1.04^7. . . . . .
5. Calculate $2^2 \times 2^3$. . . . . .

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today. Before the activity, have students rephrase the key words: build an exponential model and identify the parameters.

Investigation

An app has 12,000 users at launch and its usage increases by 9% per month. Construct $f(t)$, calculate $f(6)$, then determine numerically when it will exceed 25,000 users.



Written summary

A constant relative change with multiplier q is modelled by $f(t) = V_0 q^t$. V_0 is the initial initial and q the coefficient per unit of time.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Simplify $3^{x+5} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 5 \times 1{,}05^x$ and $g(x) = -5 \times 0{,}8^x$.
3. An initial value of 1650 increases by 5% per year. Give the model and calculate the value after 6 years.

References in the textbook

Extensions: Calao, p. 50, nos. 20 to 25, then p. 53, no. 41. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the textbook's digital resources.

Calao textbook · p. 50, Exercise 25

Atmospheric pressure is 1,013 hPa (hectopascal) at sea level and decreases by 12% each time one climbs 1,000 m. This is exponential decay.
It can be modelled by a function P of altitude h in thousands of metres satisfying: $P(h) = ka^h$.

1. Determine the constants k and a.
2. Calculate the pressure at an altitude of 5,500 m, to the nearest hPa.

Independent work

Redo practice exercise 2 with the following data: replace the initial value with 185 and the rate or common ratio with 3. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 14 — Chapter review and assessment

Exponential functions

12/11/2026 • B : 1 h – C : 2 h

Objectives of the session

· Use what was learnt in the chapter.

· Write up a model.

· Correct an error.

Fluency · 6 to 8 minutes, without a calculator

1. Simplify $3\quad x+3-x+2$. .....
2. Is the function $x \mapsto 0{,}7^x$ increasing? .....
3. Compare $1.04^{\ 3}$ and 1.04^7. .....
4. Calculate $2^{\ 4} \times 2^3$. .....
5. Write $5^{\ -2}$ in fraction form .....

Reminder and prerequisites

Reread the last written summary and be able to carry out the basic calculations used today.
Before the activity, have students rephrase the key words: use what the chapter has taught and write up a model.

Investigation

From four pieces of student work, identify and correct: rate/multiplier confusion, incorrect property of powers, error in the direction of variation and conclusion without a unit.

Written summary

To validate an exponential model: identify V_0 and q, calculate with units, check the order of magnitude and interpret the result.

Worked example

Take the data from the activity and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then answer sentence. The model answer appears in the teacher's edition.

Practice · essentials

1. Simplify $3^{\ x+6} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 6 \times 1{,}06^x$ and $g(x) = -6 \times 0{,}8^x$.
3. An initial value of 1700 increases by 6% per year. Give the model and calculate the value after 6 years.

References in the textbook

Extensions: Calao, p. 54, nos. 47 to 58. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and the digital resources of the textbook.

Calao textbook · p. 55, Exercise 59

1. A share falls by 20% one year, then by 80% the following year. What is the average annual percentage decrease?

2. What is the average rate of change corresponding to a 60% increase followed by a decrease of 60%?

Independent work

Redo practice exercise 3 with the following data: replace the initial value with 190 and the rate or common ratio with 4. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.